

**Name:**

**DNI/Passport:**

**MASTER IN DATA SCIENCE (FIB-UPC).**  
**ACADEMIC YEAR 23-24 Q1 – MIDTERM EXAM**  
**Statistical Inference and Modelling (SIM).**

**Date: 3/Nov/2023 15:30-17:30 h**

**Classrooms A5S102 - Group 11, A5S111-Group 12**

|                        |                                                                                                                                                                                                   |
|------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Professor:</b>      | Lidia Montero and Josep Franquet                                                                                                                                                                  |
| <b>Rules for quiz:</b> | Internet access is required, emailing and chatting is strictly forbidden.<br>Mobile phones should be switched off. R documents in folder on the ATENEA Virtual Campus are allowed during the exam |
| <b>Duration:</b>       | 2h 00 min                                                                                                                                                                                         |
| <b>Marks:</b>          | Before 14/Nov/23 Subject ATENEA WEB site.                                                                                                                                                         |
| <b>Open Office:</b>    | November 15 <sup>th</sup> 2023 at 20h after class. Email requests also allowed.                                                                                                                   |

**Problem 1: All questions account for 1 point (you have to answer all of them)**

Dataset for this exercise can be found on stata website (<http://www.stata.com/texts/eacsap/>) and it was used to produce the output in the book *Econometric Analysis of Cross Section and Panel Data* by Jeffrey Wooldridge. There are 680 observations about class attendance and results on assignments.

| variable name | storage type | display format | value label | variable label               |
|---------------|--------------|----------------|-------------|------------------------------|
| attend        | byte         | %8.0g          |             | classes attended out of 32   |
| termgpa       | float        | %9.0g          |             | GPA for term                 |
| priGPA        | float        | %9.0g          |             | cumulative GPA prior to term |
| ACT           | byte         | %8.0g          |             | ACT score                    |
| final         | byte         | %8.0g          |             | final exam score             |
| atndrte       | float        | %9.0g          |             | percent classes attended     |
| hwrtte        | float        | %9.0g          |             | percent homework turned in   |
| frosh         | byte         | %8.0g          |             | =1 if freshman               |
| soph          | byte         | %8.0g          |             | =1 if sophomore              |
| skipped       | byte         | %9.0g          |             | number of classes skipped    |
| stndfnl       | float        | %9.0g          |             | (final - mean)/sd (target)   |

A descriptive statistic summary() based on R shows:

```
> summary(attend)
      attend      termgpa      priGPA      ACT      final
Min.   : 2.00   Min.   :0.000   Min.   :0.857   Min.   :13.00   Min.   :10.00
1st Qu.:24.00   1st Qu.:2.138   1st Qu.:2.190   1st Qu.:20.00   1st Qu.:22.00
Median :28.00   Median :2.670   Median :2.560   Median :22.00   Median :26.00
Mean   :26.15   Mean   :2.601   Mean   :2.587   Mean   :22.51   Mean   :25.89
3rd Qu.:30.00   3rd Qu.:3.120   3rd Qu.:2.943   3rd Qu.:25.00   3rd Qu.:29.00
Max.   :32.00   Max.   :4.000   Max.   :3.930   Max.   :32.00   Max.   :39.00

      atndrte      hwrtte      frosh      soph      skipped
Min.   : 6.25   Min.   :12.5   Min.   :0.0000   Min.   :0.0000   Min.   : 0.000
1st Qu.:75.00   1st Qu.:87.5   1st Qu.:0.0000   1st Qu.:0.0000   1st Qu.: 2.000
Median :87.50   Median :100.0   Median :0.0000   Median :1.0000   Median : 4.000
Mean   :81.71   Mean   :87.9   Mean   :0.2324   Mean   :0.5765   Mean   : 5.853
3rd Qu.:93.75   3rd Qu.:100.0   3rd Qu.:0.0000   3rd Qu.:1.0000   3rd Qu.: 8.000
Max.   :100.00   Max.   :100.0   Max.   :1.0000   Max.   :1.0000   Max.   :30.000
      NA's      : 6.0

      stndfnl
Min.   :-3.30882
1st Qu.: -0.78782
Median : 0.05252
Mean   : 0.02966
3rd Qu.: 0.68277
Max.   : 2.78361
```

Selected numeric target is **stndfnl** (standardized final exam score) and let freshman status be the qualitative target. Prepare your dataset to represent factors in a suitable way.

1. Determine thresholds for mild and severe outliers in the target. Are there any outliers? Indicate observation id's and atypical values. **Do not take any action.**
2. Use an imputation method to address missing data in the dataset. Validate imputation results.
3. Are there any multivariate outliers in the dataset? Indicate how many and which at 99% confidence. Explain what they seem to share in common. **Do not take any action.**
4. Analyze the profile of the numeric target (**stndfnl**) using a suitable profiling method. A detailed explanation of the procedure outcome is requested.
5. Analyze the profile of the binary target (**frosh**) using a suitable method. A detailed explanation of the procedure outcome is requested.
6. Is there variance homogeneity in the **stndfnl** target groups defined by frosh and soph classes, one by one and when applied simultaneously? *Hint: You have to define a new factor.*
7. Mean **stndfnl** target can be considered to be the equal across groups defined by frosh target? Use a two.sided test at 1% significance level and indicate the confidence interval for freshman target population mean. Indicate whether equal variances and normal distribution of stndfnl hypothesis hold in the population.
8. State and test one.sided hypothesis to assess whether **stndfnl** is less for freshman than the rest at 1% significance level. Indicate and justify a 95% confidence interval for freshman target population mean.
9. The standard deviation of **stndfnl** in the freshman group should not exceed 1. For the sample in the freshman group in your dataset, calculate the deviation of **stndfnl** assuming that normal assumption holds. State and include any assumptions needed to test at the 1% significance level whether population standard deviation is larger than 1 in the freshman group. Figure out the 99% upper threshold for **stndfnl** in the freshman population standard deviation.
10. Determine a 99% confidence interval for the population proportion of neither freshman, nor sophomore. Test the null hypothesis that selecting a unit neither freshman, nor sophomore has a population probability greater than 0.25.
11. Test the null hypothesis that the proportion of freshman group and the population proportion of neither freshman, nor sophomore is the same at 1% significance level.

Hint:

CI for Difference in Two Sample Means – Pooled Std.Deviation

- If we assume that the standard deviation of the two populations is the same, then we can use the pooled standard error to estimate it:

$$s_p = \sqrt{\frac{(n_1 - 1)s_1^2 + (n_2 - 1)s_2^2}{n_1 + n_2 - 2}}$$

- And then the CI becomes:

$$CI = \bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2, df} \times s_p \sqrt{\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}$$

- Note that  $df = n_1 + n_2 - 2$

CI for Difference in Two Sample Means

- CI for  $\mu_1 - \mu_2$

$$CI = \bar{x}_1 - \bar{x}_2 \pm t_{\alpha/2, df} \times \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

$$df = \frac{\left(\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}\right)^2}{\frac{1}{n_1 - 1} \left(\frac{s_1^2}{n_1}\right)^2 + \frac{1}{n_2 - 1} \left(\frac{s_2^2}{n_2}\right)^2}$$

CI for Difference in two Population Proportions

$$\left( \hat{p}_1 - \hat{p}_2 \pm z_{\alpha}^* \sqrt{\frac{\hat{p}_1(1 - \hat{p}_1)}{n_1} + \frac{\hat{p}_2(1 - \hat{p}_2)}{n_2}} \right)$$

Let  $X_1, X_2, \dots, X_n$  be a random sample from a normal distribution with unknown mean  $\mu$  and unknown variance  $\sigma^2$ . The quantity

$$X^2 = \frac{(n - 1)S^2}{\sigma^2}$$

has a chi-square distribution with  $n - 1$  degrees of freedom, abbreviated as  $\chi_{n-1}^2$ . In